

Vulnerability Assessment Report

1st January 20XX

System Description

The server hardware consists of a powerful CPU processor and 128 GB of memory. It runs on the latest version of the Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment covers a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The data center is one of the organization’s most important assets, as it is responsible for managing and storing all customers’ and employees’ confidential information. If the data center suddenly stops operating or is attacked, customer data could be exploited or exposed, causing financial losses and damage to the organization’s reputation. Therefore, assessing the vulnerabilities of the data center is an important security step. This process helps make the system more stable and better protected against both external and internal threats.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	Obtain sensitive information via exfiltration	1	3	3
Employee	Alter/Delete critical information	2	3	6
Hacker	Conduct Denial of Service (DoS) attacks or obtain confidential information	3	3	9

Approach

The threat events coming from hackers and competitors are mainly related to the fact that the database is open to the public. Because of this, a hacker could perform a Denial of Service (DoS) attack, while a competitor or attacker could infiltrate the database and obtain sensitive information, damaging the organization's reputation. Another important source of threat is the employees themselves. Since all employees work remotely from different locations around the world, it is extremely important to properly manage and monitor their access requests. This helps prevent employees from intentionally or unintentionally altering or deleting critical information.

Remediation Strategy

For the first threat source, competitors, one important security control is Public Key Infrastructure (PKI), an encryption framework that secures the exchange of information online and helps prevent unauthorized users from obtaining sensitive data.

For the second threat source, employees, important security controls include the principle of least privilege, where employees only receive the minimum level of access necessary to perform their tasks, and separation of duties, which helps prevent system misuse.

To defend against hackers, the organization should follow network security hardening practices. This includes investing in and maintaining firewalls to monitor suspicious connections from external networks, applying regular patch updates, and using network monitoring tools such as Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS).